

BCA Semester –III

Subject Title (Theory): Object Oriented Programming using JAVA

Course code	Type of Course	Theory / Practical	Credits	Instruction hour per week	Duration of Exam	Formative Assessment Marks	Summative Assessment Marks	Total Marks
C3BCA1T1	Paper-1	Theory	04	04	3hrs	20	80	100

Course Outcome: At the end of the course, students will be able to:

CO 1: Explain the object-oriented concepts using JAVA.

CO 2: Write JAVA programs using OOP concepts like Abstraction, Encapsulation,

CO 3: Inheritance and Polymorphism.

CO 4: Implement Classes and multithreading using JAVA.

CO 5: Demonstrate the basic principles of creating Java applications with GUI.

Unit I:

15 hrs.

Introduction to JAVA: JAVA Evolution: Java History, Java Features, How Java differs from C and C++, Java and Internet, Java and World Wide Web, Web browsers, Hardware and software requirements, Java Support Systems, Java Environment. Overview of JAVA Language: Introduction, Simple Java Program, More of Java, An Application with Two Classes, Java Program Structure, Java Tokens, Java Statements, Implementing a Java Program, Java Virtual Machine, Command Line Arguments, Programming Style. Constants, Variables, and Data Types: Introduction, Constants, Variables, Data Types, Declaration of Variables, Giving Values to Variables, Scope of Variables, Symbolic Constants, Type Casting, Getting Values of Variables, Standard Default Values. Operators and Expressions: Introduction, Arithmetic Operators, Relational Operators, Logical Operators, Assignment Operators, Increment and Decrement Operators, Conditional Operators, Bitwise Operators, Special Operators, Arithmetic Expressions, Evaluation of Expressions, Procedure of Arithmetic Operators, Type Conversion and Associativity, Mathematical functions. Decision Making and Branching: Introduction, Decision Making with if statement, simple if statement, if...else statement, Nesting of if.... else statements, the else if Ladder, the switchstatement, the?: Operator. Decision Making and Looping: The while statement, The do statement, The for statement, Jumps in Loops, Labelled Loops.

Unit II:

15 hrs.

Classes, Arrays, Strings and Vectors: Classes, Objects and Methods: Introduction, Defining a class, Adding Variables, Adding Methods, Creating Objects, Accessing class members, Constructors, Methods Overloading, Static Members, Nesting of Methods. Inheritance: Extending a class, Overriding Methods, Final Variables and Methods, Finalizer Methods, Abstract Methods and Classes, Visibility Control. Arrays, Strings and Vectors: Arrays, One-dimensional Arrays, Creating an array, Two-dimensional Arrays, Strings, Vectors and Wrapper Classes.

Unit III:**15 hrs.**

Interfaces, Packages, and Multithreaded Programming: Interfaces: Multiple Inheritances: Introduction, Defining Interfaces, Extending Interfaces, Implementing Interfaces, Accessing Interface Variables. Packages: Putting Classes together: Introduction, Java API Packages, Using System Packages, Naming Conventions, Creating Packages, Accessing a Package, Using a Package, Adding a Class to a Package, Hiding Classes. Multithreaded Programming: Introduction, Creating Threads, Extending the Thread Class, Stopping and Blocking a thread, Life Cycle of a thread, Using Thread Methods, Thread Exceptions, Thread Priority, Synchronization, Implementing the Runnable Interface.

Unit IV:**15 hrs.**

Managing Exceptions, Applet Programming: Managing Errors and Exception: Introduction, Types of Errors, Exceptions, Syntax of Exception Handling Code, Multiple Catch Statements, Using Finally Statement, Throwing Our Own Exceptions, Using Exceptions for Debugging. Applet Programming: Introduction, How Applets Differ from Applications, Preparing to Write Applets, Building Applet Code, Applet Life Cycle, Creating an Executable applet, Designing a Web Page, Applet Tag, Adding Applet to HTML File, Running the Applet, More about Applet Tag, Passing Parameters to Applets, Aligning the Display, More About HTML Tags, Displaying Numerical Values, Getting Input from the user. Graphics Programming, Input/Output: Graphics Programming: Introduction, The Graphics class, Lines and rectangles, circles and ellipses, Drawing Arcs, Drawing Polygons, Line Graphs, Using Control Loops in Applets, Drawing Bar Charts.

Text Books:

1. Shishir Gundavaram, CGI Programming on the World Wide Web, O'Reilly and Associates, (1996). (Chapter 1-7)
2. E. Balagurusamy, Programming with JAVA, A Primer, 2nd Edition. TMH (1999), (Chapter 2-16)

Reference Books:

1. Thomas Boutel, CGI Programming in C and Perl, Addison—Wesley, (1996).
2. Jeffrey Dwight et al, Using CGI, (Second Edition), Prentice Hall, India, (1997).
3. Darrel Ince & Adam Freeman, Programming the Internet with Java, Addison—Wesley, (1997).
4. Ken Arnold & James Gosling, The Java Programming Language, Addison—Wesley, (1998).
5. Patrick Naughton & Herbert Schildt, JAVA 2: The Complete Reference, 3rd Edition, TMH, (1999).

Formative Assessment for Theory	
Assessment Occasion/ type	Marks
Internal Assessment Test 1	05
Internal Assessment Test 2	05
Assignment	10
Total	20
<i>Formative Assessment as per guidelines</i>	